EXOPLANET DETECTION USING WAVELETS TRANSFORMS AND CONVOLUTIONAL NEURAL NETWORKS

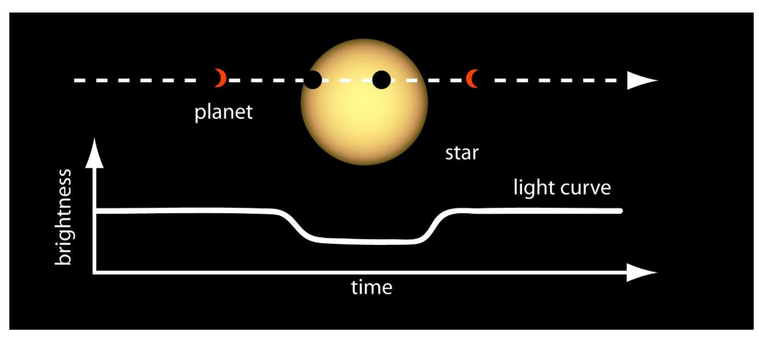
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INTRODUCTION

Iconic missions such as Kepler and TESS have collected massive volumes of photometric data to detect extrasolar planets via the transit method, where the fractional dimming of stellar flux (δ) is governed by the geometric ratio of the planet (R_p) and star (R_*) radii:

$$\delta = \left(\frac{R_p}{R_*}\right)^2$$

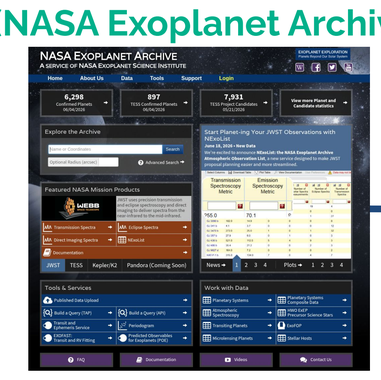
The main challenge lies in distinguishing these subtle morphological dips from systematic interference or stellar false positives (such as eclipsing binaries or intrinsic variability).



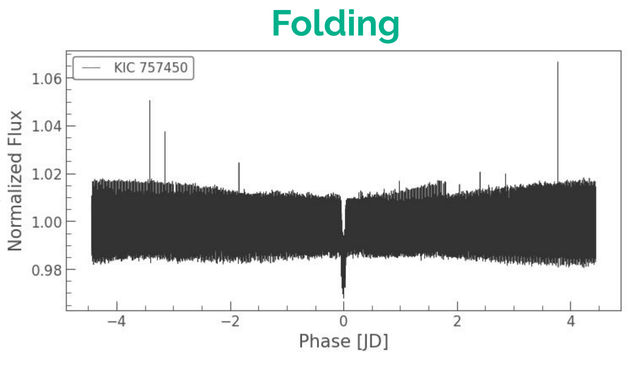
Objective:
To develop and evaluate a 2D CNN architecture optimized via wavelet transforms for the automated and robust classification of noisy light curves from the Kepler mission.

METHODOLOGY

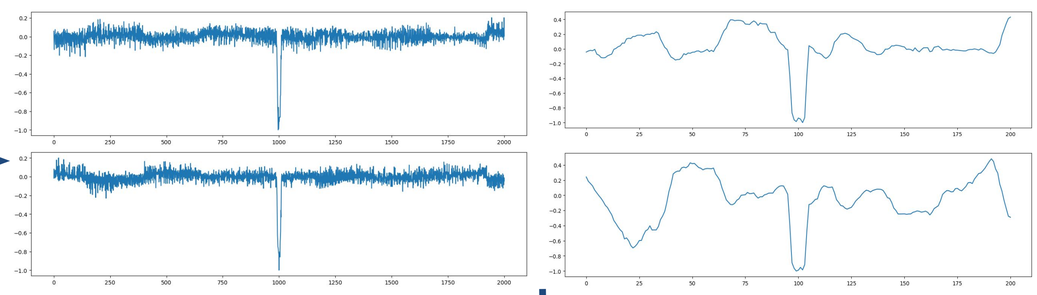
1. Data Download (NASA Exoplanet Archive)



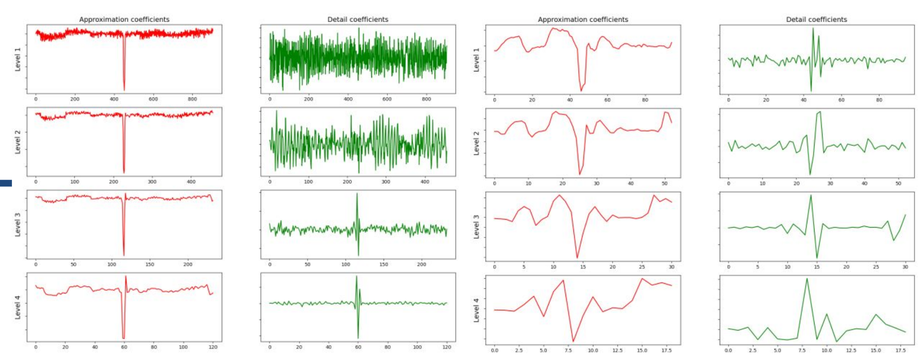
2. Normalization and Phase Folding



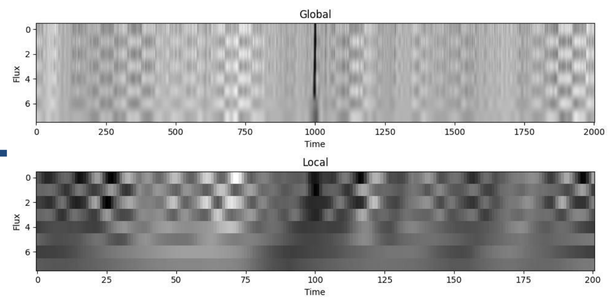
3. Dual-Scale Binning (Global/Local)



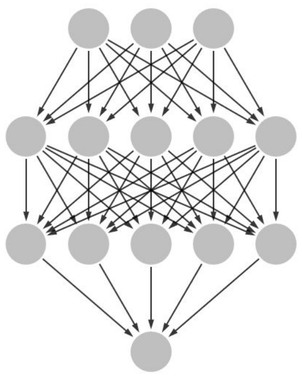
4. Wavelet Transforms



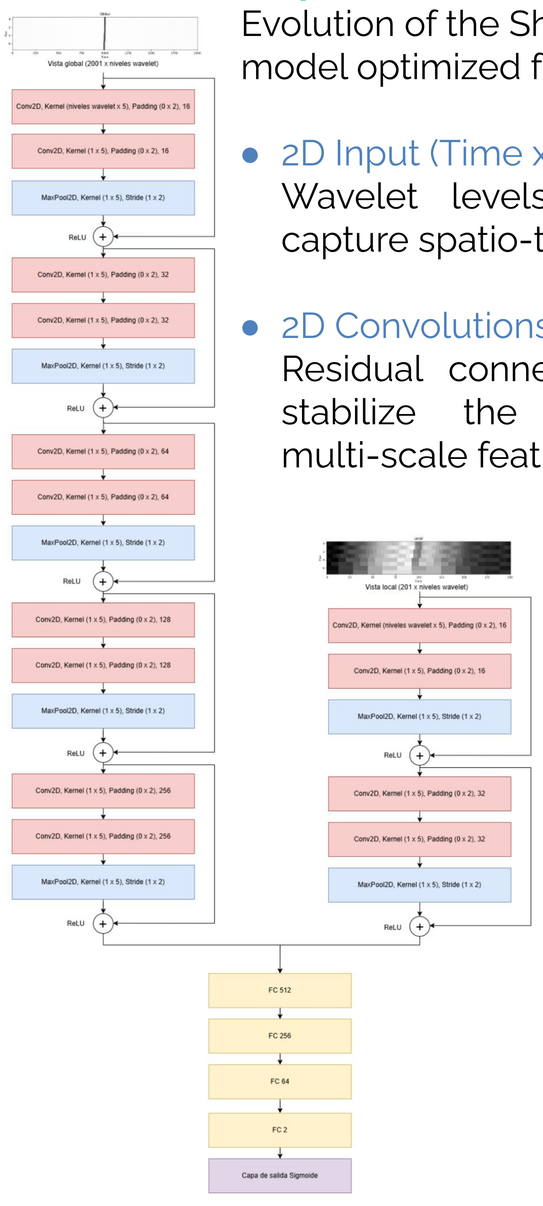
5. 2D Composition



6. Classification



ARCHITECTURE AND RESULTS



Proposed Model:

Evolution of the Shallue & Vanderburg (2018) model optimized for multi-scale data:

- 2D Input (Time x Scala): Wavelet levels stacked as images to capture spatio-temporal patterns.
- 2D Convolutions & Skip Layers: Residual connections (ResNet-style) that stabilize the gradient and combine multi-scale features.
- Dense Bottleneck: A bottleneck in the fully connected layers to force feature abstraction and prevent overfitting.

Training Strategy and Real-World Data:

- Real Dataset: ~7,500 Kepler curves (Confirmed and False Positives). Candidates were strictly excluded from training to avoid label noise.
- No Alterations: No Data Augmentation or simulated curves were applied, evaluating model performance strictly against the original mission curves.

Test Set Performance:

- The proposed model achieves a higher F1-score for smooth and symmetric wavelet families (Meyer, Symlet, B3-Spline), offering maximum synergy with 2D convolutions.



Measuring Real-World Temporal Robustness

100%
PRECISION ON REAL CANDIDATES

- Predictive Simulation: Strict training with the catalog available in 2022 and "blind" inference on candidates from that period.
- Future Validation: Cross-referencing predictions with official updates consolidated in the 2025 Kepler catalog.
- Temporal Performance: Reaches an outstanding 76% F1-Score, driven by a flawless 100% Precision (zero false alarms among transitioned candidates) and 61% Recall (sensitivity).

CONCLUSIONS AND FUTURE LINES

1. 2D Synergy: Stacking wavelet approximations into 2D matrices enables the extraction of hierarchical patterns inaccessible to classical 1D models.
2. Critical Morphology: Smooth and symmetric wavelets (Meyer, B3-Spline) mitigate numerical artifacts and preserve the true geometric profile of the transit.
3. Temporal Validation: The 76% F1-Score in the candidates test (2022 vs 2025) demonstrates high real-world reliability, remarkably backed by a flawless 100% Precision in positive exoplanet alerts.

Future Lines:

1. End-to-End Evolution: Preliminary tests using advanced deep models directly on raw data show a powerful breakthrough path, currently raising candidate metrics even further.
2. On-Field Validation: Conducting follow-up observations of candidates to directly evaluate model performance.

BIBLIOGRAPHY

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